THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **The Wonderful World of Science, Diversity in the Living World, Mindful Eating: A Path to a Healthy Body, Exploring Magnets** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** the container and the flame also differ along with the volume	Volume of water is a variable.
2	**(a)** a grass	Grasses show these features.
3	**(d)** starch and fat	Both tests are positive.
4	**(a)** P and R are magnets, Q is magnetic material	Repulsion is seen only between magnets.
5	**(a)** both depend on hot weather	A third factor links them.
6	**(c)** helps judge distance for hunting	Forward eyes give depth perception.
7	**(b)** they contain much sugar and few nutrients	Empty calories.
8	**(a)** iron is magnetic and copper is not	Only iron is attracted.
9	**(d)** compare the methods and repeat carefully	Differences in method may explain.
10	**(c)** many plants will fail to pollinate	Pollination would fall.
11	**(b)** fat	Protein test.
12	**(a)** always in the same direction with the same pole of a magnet	Opposite pole forms at the lifting end.
13	**(a)** Both A and R are true and R is the correct explanation of A.	A hypothesis is a testable explanation, and testing it by an experiment is exactly how we check it.
14	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but R is a general pattern and does not explain why wheat has fibrous roots.
15	**(a)** Both A and R are true and R is the correct explanation of A.	Growth and repair of tissues is why proteins are called body-building nutrients.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Two factors differ: light and water (1); We cannot tell which factor caused the difference (1).	1+1
17	Green stem carries out photosynthesis (1); Spines reduce water loss (1).	1+1
18	It contains starch (1); It also contains fat (1).	1+1
19	The steel table disturbs the needle (1); Move it to a wooden or non-magnetic surface (1).	1+1
20	It represents the Earth in small size (1); It helps us study things too large to observe wholly (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Use same size, same wetness (1); Same place, time and temperature (1); Measure drying time for each (1).	1+1+1
22	Tree (1); Tap root goes deep (1); Net veins in leaves (1).	1+1+1
23	Lacks protein and vitamins (1); High in refined carbohydrates and fat (1); Add milk, dal, vegetables and fruit (1).	1+1+1
24	Roll bar in filings (1); Maximum cling near the ends (1); Little in the middle (1).	1+1+1
	OR: Curiosity: why do things fall (1); Observation: fall of apple and Moon's motion (1); Reasoning: same force acts on both (1).	1+1+1
25	Colour can vary within a group (1); Unrelated animals may have the same colour (1); Features like body covering and habitat are better (1).	1+1+1
	OR: Calcium for bones (1); Proteins for growth (1); Vitamins like A and D (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

26	(i) A hypothesis (1). (ii) The amount of sunlight (1). (iii) Recording the observations (or results) (1). (iv) Only one thing should change so that any difference can be blamed on sunlight (1); the plant in sunlight grew better, so plants need sunlight (1).	1+1+1+2
27	A: vitamin C, citrus fruits (1); B: vitamin A, carrot (1); C: iodine, iodised salt (1); D: vitamin D, sunlight and milk (1); E: iron, spinach (1).	1+1+1+1+1
	OR: Same type of clips (1); Same method to approach (1); Count clips lifted by each pole (1); Repeat three times (1); Compare averages (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **The Wonderful World of Science, Diversity in the Living World, Mindful Eating: A Path to a Healthy Body, Exploring Magnets** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** south-seeking pole	The end that points south is the south-seeking pole.
2	**(b)** amount of water	Water amount varies.
3	**(b)** wide field of view to spot predators	Wide view.
4	**(b)** protein and fat	Plant foods have less protein and fat.
5	**(a)** iron only	Only iron is magnetic.
6	**(a)** rounding off numbers to look neat without noting it	Data must not be altered.
7	**(b)** it provides more food, water and shelter	Rich resources support life.
8	**(a)** X has starch, Y has fat, Z has neither starch nor fat	Match tests with nutrients.
9	**(b)** B's north repels A's north	Like poles repel.
10	**(c)** revise the hypothesis	Data must be respected.
11	**(a)** to avoid sinking in sand	Large area reduces pressure.
12	**(c)** spinach and jaggery	Iron-rich foods.
13	**(a)** Both A and R are true and R is the correct explanation of A.	The Earth's magnetic field acts on the suspended magnet and aligns it, so R explains A.
14	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the use of a telescope does not explain why observation comes first.
15	**(a)** Both A and R are true and R is the correct explanation of A.	The smooth, tapering body reduces resistance in water, so R explains A.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	It breathes air through lungs (1); It gives birth to young and feeds milk (1).	1+1
17	Lacks carbohydrates and fibre (1); Lacks many vitamins (1).	1+1
18	No (1); Heat weakens the magnet (1).	1+1
19	Example: plastic waste (1); Science develops recycling or biodegradable materials (1).	1+1
20	Water supports them (1); Flexible stems bend with the flow of water without breaking (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Permanent magnet is present (1); A coil moves in its field (1); Produces sound (1).	1+1+1
22	A factor that can change (1); Temperature of water (1); Amount of sugar and stirring (1).	1+1+1
23	Beak shape matches food (1); Woodpecker has strong beak for wood (1); Hummingbird has thin beak for nectar (1).	1+1+1
24	Fruits and nuts (1); Roasted chana (1); Avoid sugary packaged items (1).	1+1+1
	OR: Iron nails are magnetic (1); Brass screws are not magnetic (1); The magnet picks up only the nails (1).	1+1+1
25	P vs Q tests light (1); P vs R tests water (1); Only one factor differs in each comparison (1).	1+1+1
	OR: Duck: webbed feet (1); Fish: fins and tail (1); Fish: streamlined; duck: oily feathers (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) Mango (1). (ii) Pumpkin (1). (iii) Herb: small with soft, green stem; shrub: bushy with a hard stem branching near the base (1). (iv) A creeper spreads on the ground with a weak stem (1); a climber climbs up a support with tendrils or other parts; example: grapevine (1).	1+1+1+2

27	Cereal for energy (1); Dal for proteins (1); Vegetables for vitamins (1); Curd for calcium (1); Salad and pickle for fibre and flavour (1).	1+1+1+1+1
	OR: Roll magnet in filings (1); Filings gather at the ends (1); Fewer in middle (1); Count or compare quantity (1); Poles are strongest (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **The Wonderful World of Science, Diversity in the Living World, Mindful Eating: A Path to a Healthy Body, Exploring Magnets** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** cereals with pulses	Combination improves protein quality.
2	**(d)** Earth's magnetism slowly magnetises it	Induction by Earth's field.
3	**(a)** Why do wet clothes dry faster on a windy day?	It looks for a cause.
4	**(b)** animals use oxygen produced by plants	Plants give oxygen.
5	**(c)** easily digestible, nutritious food	Recovery needs nutrients.
6	**(a)** unlike poles attract	Earth's north is magnetic south.
7	**(c)** Does temperature affect how fast sugar dissolves?	Testable questions have measurable answers.
8	**(b)** thick fur, fat layer, white colour	Cold adaptation.
9	**(c)** vitamin A, night blindness	Vitamin A deficiency affects dim-light vision.
10	**(a)** a magnet or iron object nearby	Local magnetic influence.
11	**(c)** curiosity about everyday things can lead to great discoveries	Simple tools with careful thinking can lead to discoveries.
12	**(c)** prop roots	They support the tree.
13	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the food source does not explain the disease.
14	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true, but R does not explain why iron is attracted.
15	**(c)** A is true but R is false.	A is true because repeating avoids mistakes; R is false, a single trial can be wrong.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	For growth of tissues (1); For repair of body cells (1).	1+1
17	A pin can be moved through paper by a magnet (1); Non-magnetic sheets do not block force (1).	1+1
18	It shows what happens without the change (1); Example: unfertilised plants compared to fertilised ones (1).	1+1
19	No (1); Bats and insects also fly (1).	1+1
20	The skin makes vitamin D in sunlight (1); Vitamin D helps the body use calcium (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Dal adds protein (1); Dal adds minerals (1); Together they give a more balanced intake (1).	1+1+1
22	Gave direction at sea (1); Not dependent on stars (1); Safer voyages (1).	1+1+1
23	Hand is unreliable (1); It depends on earlier experience (1); A thermometer gives exact readings (1).	1+1+1
24	Stores fat (1); Fat gives energy when food is scarce (1); Body fat in one place keeps the rest cooler (1).	1+1+1
	OR: Vitamins get destroyed by air and light (1); Wash before cutting (1); Cut just before eating (1).	1+1+1
25	Hang a bar magnet freely (1); It settles N-S (1); Repeated turns give same result (1).	1+1+1
	OR: New experiments contradict it (1); A better explanation covers more facts (1); Example: Earth at centre idea was given up (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

26	(i) Protein (1). (ii) Vitamins (and minerals) (1). (iii) The ingredients and nutrients in the food (and its date of expiry) (1). (iv) No single food gives all the nutrients (1); a mix gives carbohydrates, proteins, vitamins and minerals in balance, while junk food has too much fat, sugar and salt (1).	1+1+1+2
27	Breakfast: milk, fruit, paratha (1); Lunch: roti, dal, vegetable, curd (1); Snack: sprouts or fruit (1); Dinner: light khichdi with vegetables (1); Avoid chips, soda and excess sweets (1).	1+1+1+1+1
	OR: Test each pair for attraction (1); Copper is attracted by none (1); The magnet may repel another bar's end (1); Iron is attracted by the magnet but not repelled (1); All three identified (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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